SYBCA-AI (Semester III)

303 Machine Learning - Mini Project

DIABETES PREDICTION SYSTEM USING MACHINE LEARNING

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Certificate

This is to certify that **Yashvi Dobariya,Charvi Beladiya** (Roll No. 556, 575) has completed the Machine Learning mini project titled **Diabetes Prediction System Using Machine Learning** as a part of SYBCA-AI (Semester III), subject 303 Machine Learning.

Faculty Signature: _____

Date: _____

1. Problem Statement

Diabetes is a serious long-term health condition, and early detection helps patients start treatment sooner and avoid complications. Manual screening takes time and depends on expert judgement. This project builds a supervised binary classification model that predicts whether a patient is likely to have diabetes (1) or not (0) from medical and demographic attributes such as age, BMI, blood pressure, glucose level and family history.

2. Objectives

- Study the diabetes dataset and identify numerical, categorical, missing and duplicate values.
- Perform visual and automated Exploratory Data Analysis (EDA).
- Preprocess the data: handle missing values and outliers.
- Select the features, split the data into training and testing sets.
- Build a Logistic Regression classification model.
- Evaluate it using Accuracy, Precision, Recall, F1 Score and the Confusion Matrix.
- Improve the model using Cross Validation and Regularization (L1/L2), and compare the results.

3. Dataset Description

Source: Kaggle - Diabetes dataset (CSV downloaded by the students). The dataset contains **1000** records and **13** columns. The target variable is **Diabetes** (0 = not diabetic, 1 = diabetic). Class 0 has 78 records (7.8%) and class 1 has 922 records (92.2%).

The identifier column(s) **Patient_ID** were dropped because an ID has no predictive meaning.

Column	Type	Role
Age	Numerical	Feature (independent)
Gender	Categorical	Feature (independent)
BMI	Numerical	Feature (independent)
Blood_Pressure	Numerical	Feature (independent)
Glucose_Level	Numerical	Feature (independent)
HbA1c	Numerical	Feature (independent)
Insulin_Level	Numerical	Feature (independent)
Physical_Activity	Categorical	Feature (independent)
Smoking_Status	Categorical	Feature (independent)
Alcohol_Consumption	Categorical	Feature (independent)
Family_History	Numerical	Feature (independent)
Hypertension	Numerical	Feature (independent)
Diabetes	Binary (0/1)	Target (dependent)

The dataset has 8 numerical and 4 categorical feature columns. It satisfies the minimum requirement of 300 records and 5 or more features, and it is a real-world dataset.

4. Data Preprocessing

Missing values found in the raw data: **334**. Duplicate rows found: **0**.

Missing values after treatment: **0**. Numerical missing values are filled with the median and categorical missing values with the mode.

Outliers were detected with the IQR method (values below Q1 - 1.5 x IQR or above Q3 + 1.5 x IQR) and capped at those limits, so no records were lost.

Column	Outliers before	Outliers after
Age	0	0
BMI	8	0
Blood_Pressure	0	0
Glucose_Level	0	0
HbA1c	0	0
Insulin_Level	0	0

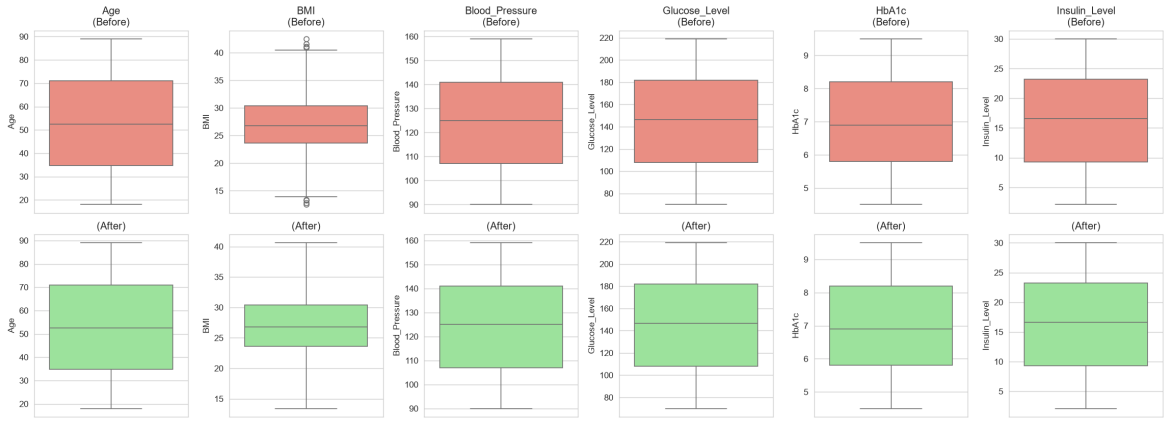


Figure 1: Box plots before and after outlier treatment.

5. Exploratory Data Analysis

The analysis includes histograms, box plots, a target pie chart, categorical bar charts and a correlation matrix. An automated EDA report (EDA_Report.html) is also generated with Sweetviz.

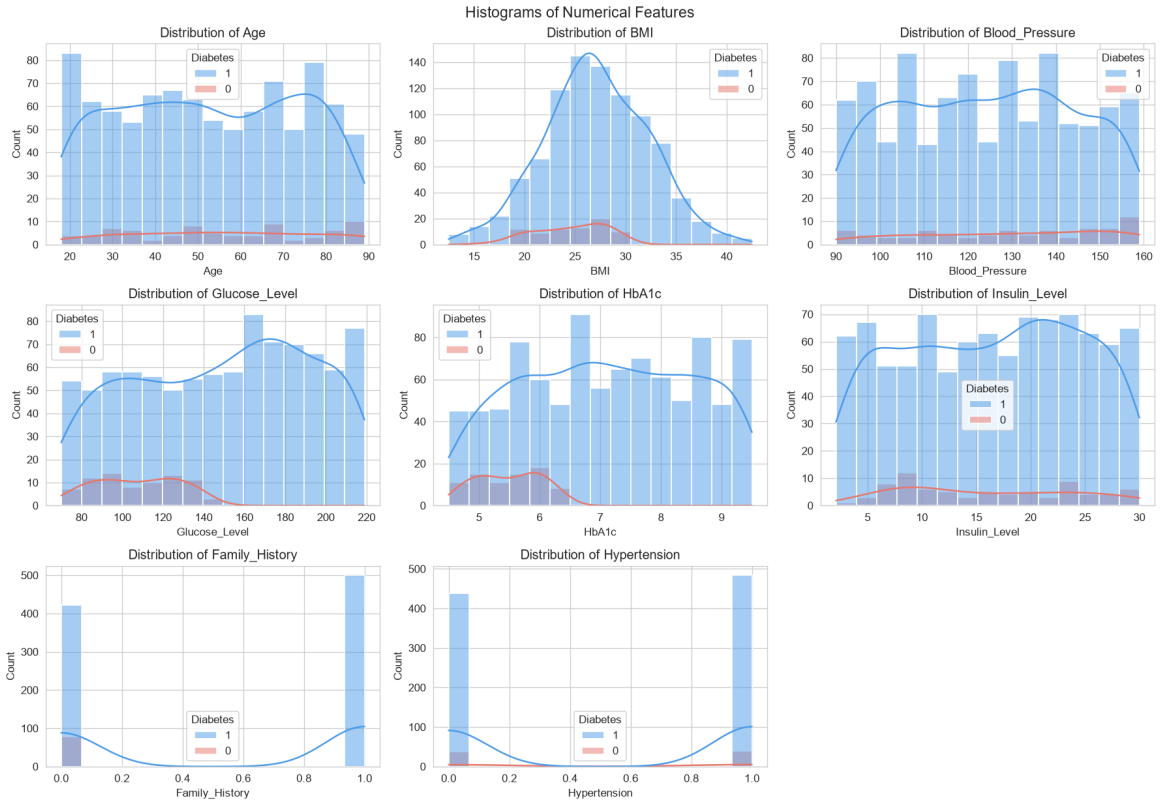


Figure 2: Histograms of numerical features by class (with KDE).

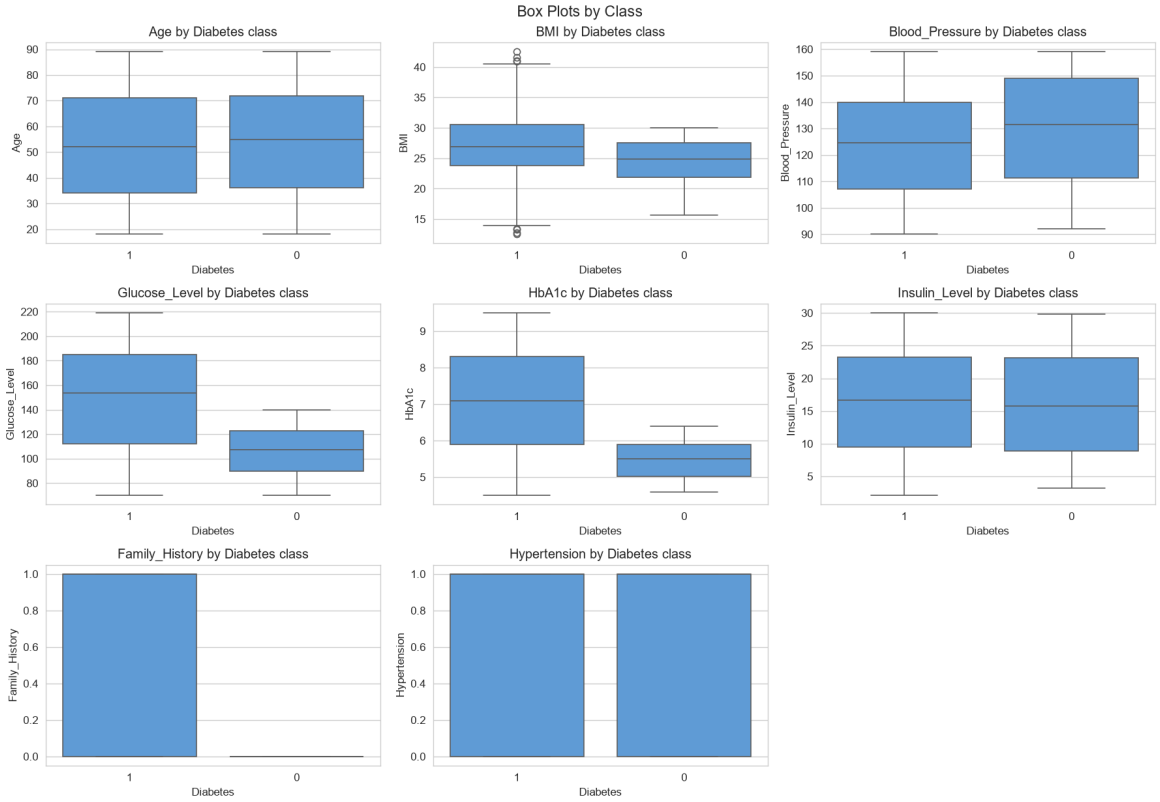


Figure 3: Box plots of numerical features by class.

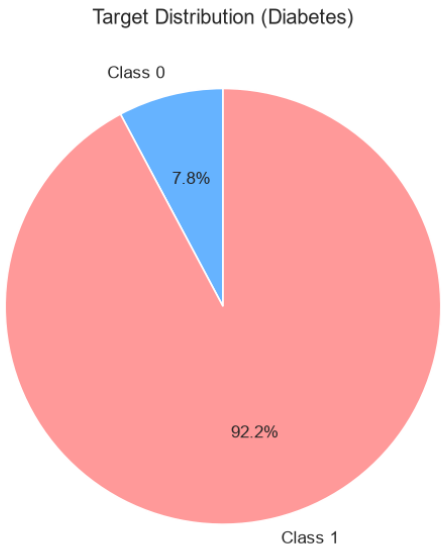


Figure 4: Distribution of the target variable (Diabetes).

The target is imbalanced: 7.8% of patients belong to class 0 and 92.2% to class 1.

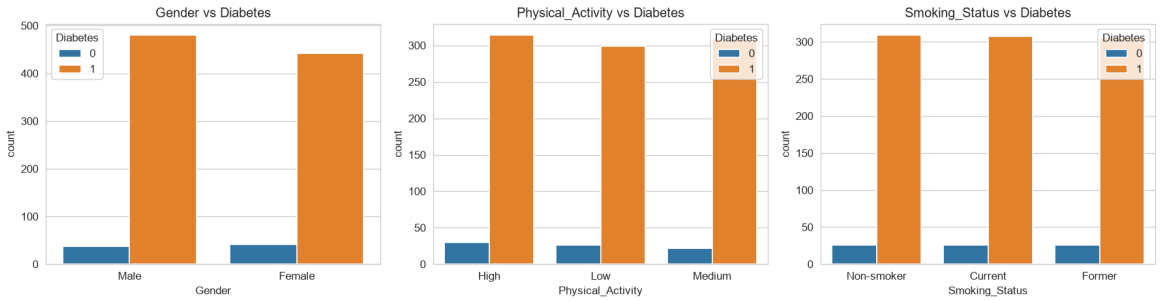


Figure 5: Categorical features versus the target.

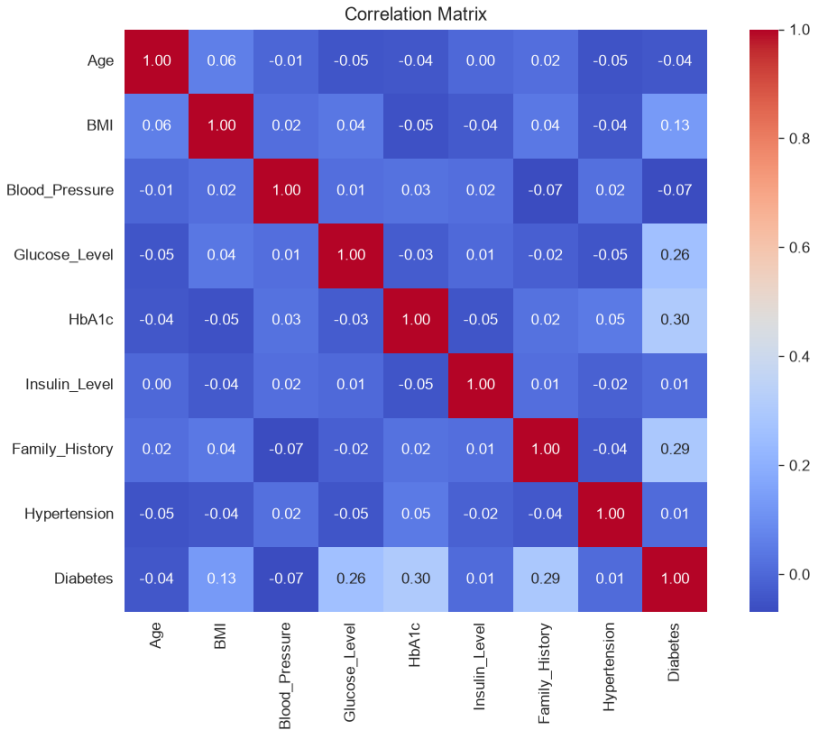


Figure 6: Correlation matrix of numerical features and the target.

Numerical features most strongly correlated with the target: HbA1c (+0.30), Family_History (+0.29), Glucose_Level (+0.26).

6. Model Building (Code)

The independent variables are all columns except **Diabetes**. Categorical columns are one-hot encoded (drop_first=True), which gives **14** input features. The data is split into 80% training (**800** records) and 20% testing (**200** records) with stratification, so both sets keep the same class ratio. The features are standardized with StandardScaler and a Logistic Regression model is trained, because the target is a binary class.

```
X = df.drop(columns=["Diabetes"])
y = df["Diabetes"].astype(int)
X = pd.get_dummies(X, columns=categorical_cols, drop_first=True)

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y)

model = Pipeline([
    ("scaler", StandardScaler()),
    ("model", LogisticRegression(max_iter=1000, random_state=42)),
])
model.fit(X_train, y_train)
pred = model.predict(X_test)
```

7. Model Evaluation

The model is evaluated on the unseen test data with four classification metrics:

- Accuracy - the share of all predictions that are correct.
- Precision - of the patients predicted diabetic, how many truly are.
- Recall - of all truly diabetic patients, how many the model found.
- F1 Score - the balance between Precision and Recall.

Model	Accuracy	Precision	Recall	F1 Score
Baseline Logistic Regression	0.9600	0.9681	0.9891	0.9785
Improved (L2, C=1.0)	0.9600	0.9681	0.9891	0.9785

Confusion matrix of the baseline model: True Negative = 10, False Positive = 6, False Negative = 2, True Positive = 182. In a medical problem, Recall is very important because a missed diabetic patient (false negative) is more harmful than a false alarm. The model's Recall is 98.91%.

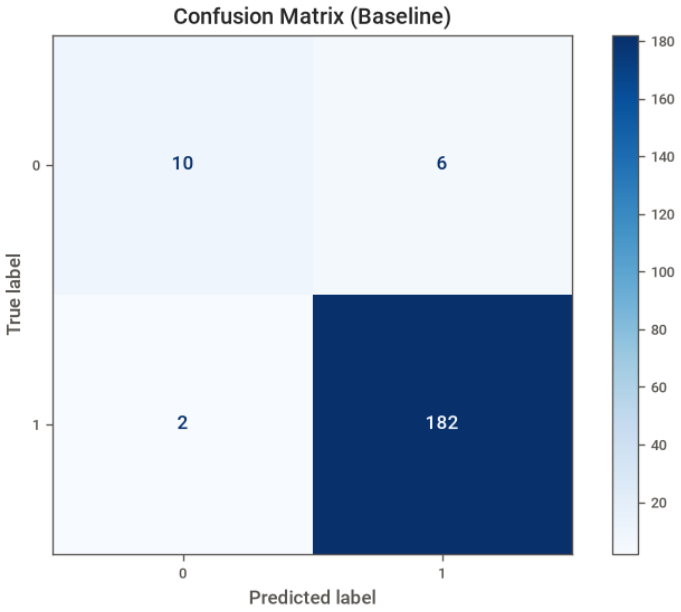


Figure 7: Confusion matrix on the test data.

Cross validation: 5-fold stratified cross validation on the training data gives a mean accuracy of 0.9525 (+/- 0.0146). **Regularization:** L1 and L2 penalties were compared for C = 0.01, 0.1, 1, 10 and 100 using the cross-validated F1 score.

Penalty	C	CV F1 score
L1	0.01	0.9597
L1	0.1	0.9678
L1	1.0	0.9745
L1	10.0	0.9723
L1	100.0	0.9723
L2	0.01	0.9597
L2	0.1	0.9678
L2	1.0	0.9752
L2	10.0	0.9716
L2	100.0	0.9723

The best setting was **L2** with **C = 1**.

8. Results

The final model reaches an accuracy of **96.00%**, precision of **96.81%**, recall of **98.91%** and an F1 score of **97.85%** on unseen test data. The improved model gives the same test result as the baseline model, which means the baseline Logistic Regression was already well suited to this data and tuning did not change the test performance.

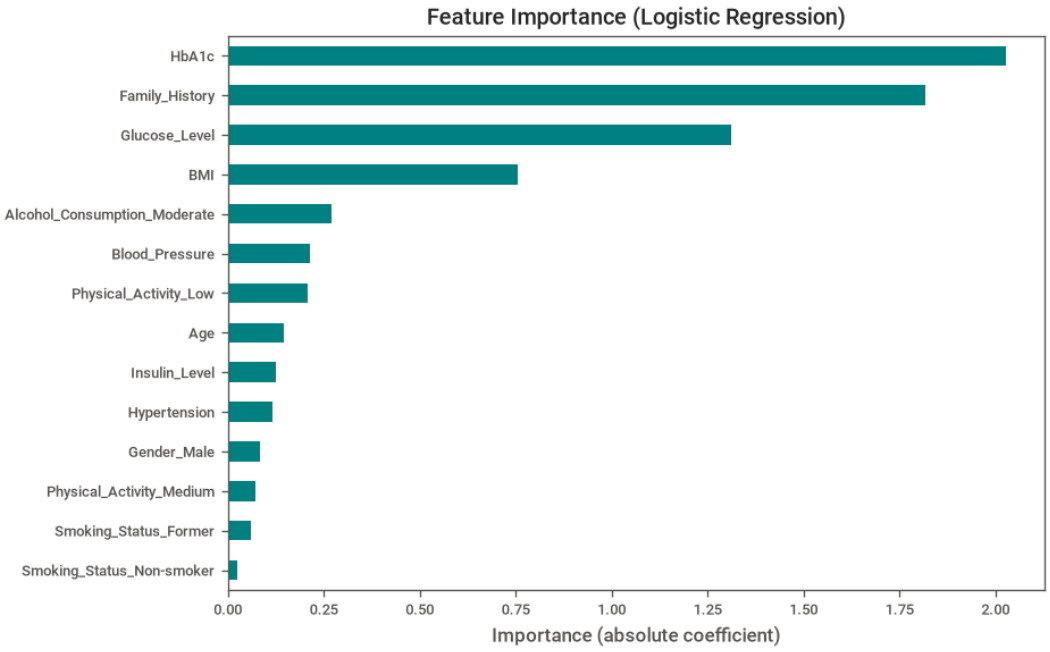


Figure 8: Feature importance (absolute coefficient).

Feature	Coefficient	Effect on diabetes risk
HbA1c	+2.026	Increases
Family_History	+1.816	Increases
Glucose_Level	+1.312	Increases
BMI	+0.757	Increases
Alcohol_Consumption_Moderate	+0.272	Increases
Blood_Pressure	-0.213	Decreases
Physical_Activity_Low	-0.209	Decreases
Age	-0.147	Decreases
Insulin_Level	+0.125	Increases
Hypertension	+0.115	Increases

A positive coefficient means the feature raises the probability of diabetes and a negative coefficient lowers it. The strongest features (largest absolute coefficient) are **HbA1c, Family_History, Glucose_Level**. Features with coefficients close to zero have very little influence on the prediction.

9. Conclusion

A Logistic Regression model was built to predict diabetes. On unseen test data it achieved an accuracy of 96.00% and an F1 score of 97.85%, with a recall of 98.91%. The complete machine learning workflow was followed: data understanding, EDA, preprocessing, feature selection, model building, evaluation and improvement with cross validation and regularization. The code uses a fixed random state and a pipeline, so the results are reproducible.

10. Future Scope

- Collect more diverse real-world patient data and test the model on it.
- Try other classifiers such as Random Forest and Gradient Boosting and compare them.
- Tune the decision threshold to increase recall for medical screening.
- Features such as HbA1c and glucose are themselves diagnostic tests, so test the model without them to predict risk earlier.
- Deploy the saved model in a simple web application (for example Streamlit).
- This is an academic model and must not be used as a medical diagnosis.

11. References

- Kaggle - Diabetes dataset (CSV downloaded by the students).
- Scikit-learn documentation - Logistic Regression, pipelines, model evaluation.
- Pandas, Matplotlib, Seaborn and Sweetviz documentation.